Software Requirements Specification (SRS)

Authoritative Functional, Interface, Quality, and Behavioral Specifications for HopeFund: Multi-Tenant NGO Donation Platform

[ACADEMIC & SYSTEMS ENGINEERING BASELINE] This specification establishes the official requirements engineering baseline for HopeFund. All system features, data dictionaries, business rules, and test procedures are strictly traceable to the specifications contained herein. In particular, financial donations are strictly simulated (is_simulated = TRUE), donation amounts accept any valid positive number greater than INR 0.00 with at most two decimal places (BR-PAY-02), public expense portals feature the mandatory self-reporting disclaimer (BR-EXP-04), and sensitive verification documents are quarantined outside the web root.

Specification Parameter	Project Metadata Details
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Specification Parameter	Project Metadata Details
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Lifecycle Baseline Status	Finalized and Approved Baseline

2. Document Information and Version Control

This section documents the formal metadata, governance history, and lifecycle audit records for the HopeFund Software Requirements Specification (SRS). Changes across iterations are tracked through permanent alphanumeric revisions, establishing end-to-end traceability from inception to final approval.

2.1 Document Governance and Administrative Details

Governance Property	Governance Value / Designation
Document Title	Software Requirements Specification (SRS) for HopeFund
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Lead Requirements Analyst	Senior Requirements Systems Analyst (SETM Team)
Quality Assurance Lead	Lead Test and Verification Engineer (SETM Team)
Target Assessment Body	Academic Capstone Review Committee and Systems Engineering Faculty

2.2 Document Revision History

The complete evolution of the requirements specification is detailed in the audit register below:

Version	Date	Description of Changes	Primary Author	Document Status
0.1.0	October 10, 2026	Initial requirements synthesis, domain elicitation, and stakeholder scope definition.	Systems Analyst	Working Draft
0.5.0	October 18, 2026	Integration of ISO/IEC 25010 quality model, mathematical ledger rules, and subsystem decomposition.	Software Architect	Technical Draft
0.9.0	October 24, 2026	Comprehensive quality audit, invariant decoupling, and reconciliation of all 34 business rules.	Systems Analyst & Architect	Candidate Baseline
1.0.0	October 25, 2026	Finalized academic baseline conforming to IEEE 29148. Verification of BR-PAY-02, BR-EXP-04, and full traceability.	Lead Systems Engineer	Approved Baseline

2.3 Stakeholder Approvals and Sign-Off

Formal sign-off by all primary engineering leads confirms that this document represents the approved baseline for design and verification:

Role / Responsibility	Designated Evaluator	Date Verified	Approval Disposition
Systems Engineering Lead	Lead Systems Architect	October 25, 2026	Approved Baseline v1.0.0
Requirements Engineering Lead	Senior Systems Analyst	October 25, 2026	Approved Baseline v1.0.0
Software Quality Assurance Lead	Lead Test Engineer	October 25, 2026	Approved Baseline v1.0.0
Academic Capstone Evaluator	Faculty Evaluation Panel	October 2026	Submitted for Evaluation

3. Table of Contents

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3.1 Document Navigational Roadmap

To assist diverse stakeholders in reviewing this specification, the following reading paths are recommended:

- Academic Evaluators: Focus on Section 4 (Introduction & Scope), Section 5 (Overall Description), Section 11 (Business Rules & Ledger Invariants), and Section 14 (Requirements Traceability Matrix).
- Software Architects: Focus on Section 5 (Subsystem Boundaries), Section 9 (Data Requirements & 3NF Schema), Section 10 (System Architecture & Major Components), and Section 12 (Error Handling & Recovery).
- Quality Assurance Engineers: Focus on Section 6 (Functional Requirements with Acceptance Criteria), Section 8 (Non-Functional Requirements), Section 13 (Use Cases), and Section 15 (Acceptance Criteria).
- NGO Administrators & Donors: Focus on Section 4.5 (Product Scope), Section 6.6 (Donations), Section 6.8 (Expense Proofs), Section 6.9 (Public Transparency), and Section 13 (Use Cases).

4. Introduction

4.1 Purpose

This Software Requirements Specification (SRS) establishes the definitive, verifiable, and authoritative requirements baseline for HopeFund: Multi-Tenant NGO Donation Platform. It defines the complete operational scope, domain capabilities, external interfaces, mathematical fund ledger invariants, and quality attributes governing the software system. This document is authored to serve as the foundational contract for academic capstone evaluation, architectural design, database normalization, and automated verification.

4.2 Scope

HopeFund is a multi-tenant web application designed to restore public trust in charitable giving by bridging the gap between donors and non-governmental organizations (NGOs) operating in India. The platform enforces institutional transparency through administrative onboarding verification gates, autonomous campaign management, transparent donation simulation, itemized expense tracking with mandatory proof bills, capacity-constrained facility visit scheduling, donor grievance redressal, and immutable audit logging.

Critical Scope Boundary: The platform implements a simulated financial engine (is_simulated = TRUE). No real banking, credit card processing, or legal currency transactions occur. All donations generate simulated PDF receipts clearly marked as non-tax-deductible educational artifacts. Furthermore, sensitive NGO founding documents and vendor expense bills are quarantined outside the public web root to guarantee absolute data privacy.

4.3 Intended Audience

This document is tailored for five primary stakeholder categories:

- Academic Evaluation Committee: Evaluates theoretical grounding, software engineering rigor, standards compliance, and methodology.
- Software Architects and Engineers: Directs modular design, database normalization to 3NF, API routing, and security boundaries.
- Quality Assurance Specialists: Guides test plan authoring, test case matrix generation, boundary condition testing, and acceptance verification.
- NGO Operations Personnel: Defines operational workflows for onboarding, campaign authoring, expense reporting, and visit scheduling.
- Platform Administrators: Establishes administrative review protocols, dispute resolution workflows, and security oversight procedures.

4.4 Definitions, Acronyms, and Abbreviations

Key software engineering and domain-specific terms used throughout this specification are formally defined below:

Term / Acronym	Full Expanded Form	Engineering and Operational Definition
3NF	Third Normal Form	A relational database normalization standard ensuring every non-key attribute depends solely on the primary key, eliminating data

Term / Acronym	Full Expanded Form	Engineering and Operational Definition
		redundancy and update anomalies.
ACID	Atomicity, Consistency, Isolation, Durability	A suite of transaction properties guaranteeing reliable database modifications even in the event of hardware or software crashes.
API	Application Programming Interface	A set of standardized protocols and endpoints allowing modular client-server communication via structured JSON payloads.
BR	Business Rule	A precise statement defining or constraining operational workflows, mathematical invariants, or access policies within the platform.
CSRF	Cross-Site Request Forgery	A web security attack forcing authenticated users into executing unauthorized requests; mitigated via cryptographically signed anti-CSRF tokens.
DFD	Data Flow Diagram	A graphical representation of data movement through information systems, depicting external entities, processes, data stores, and data flows.
ERD	Entity-Relationship Diagram	A conceptual modeling diagram illustrating database entity structures, attributes, and relational cardinalities.
FR	Functional Requirement	A specific behavioral capability or service that the platform must execute in response to inputs and user interactions.
MVC	Model-View-Controller	An architectural software pattern separating domain data logic (Model) from user presentation (View) and request mediation (Controller).
NFR	Non-Functional Requirement	A quality attribute defining performance, security, reliability, usability, or maintainability benchmarks for the system.
NGO	Non-Governmental Organization	A non-profit charitable entity registered under Indian law (Trust, Society, or Section 8 Company) seeking donor funding.
RBAC	Role-Based Access Control	An access management security model restricting user actions and page access based on assigned system roles (Admin, NGO, Donor, Guest).
RTM	Requirements Traceability Matrix	A bi-directional verification grid mapping requirements to architectural components, use cases, test cases, and database tables.
SRS	Software Requirements Specification	A formal engineering specification defining the complete behavioral, structural, and quality requirements of a software product.
STRIDE	Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, Elevation of Privilege	A threat classification model utilized to structure security requirements and defense mechanisms.
WBS	Work Breakdown Structure	A hierarchical decomposition of the total project scope into manageable, deliverable-oriented work packages.

4.5 References and Applicable Standards

This specification is formulated in rigorous accordance with international software engineering standards:

- IEEE Std 29148-2018: Systems and software engineering: Life cycle processes: Requirements engineering.
- IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications.
- ISO/IEC 25010:2011: Systems and software engineering: Systems and software Quality Requirements and Evaluation (SQuaRE): System and software quality models.
- OMG Unified Modeling Language (UML) Specification, Version 2.5.1 (Object Management Group).
- ISO/IEC/IEEE 29119:2013: Software and systems engineering: Software testing (Parts 1 through 4).
- IEEE Std 829-2008: IEEE Standard for Software and System Test Documentation.
- W3C Web Content Accessibility Guidelines (WCAG) 2.1 (Level AA Compliance).

5. Overall Description

5.1 Product Perspective

HopeFund is a completely self-contained, responsive, multi-tenant web application. It does not depend on legacy client-side systems or third-party proprietary donation networks. As a capstone platform engineering deliverable, it operates locally within standard Apache/PHP/MySQL environments (e.g., XAMPP) while architecturally prepared for zero-downtime migration to modern cloud Virtual Private Clouds (VPC).

5.2 Product Functions Overview (14 Operational Domains)

The operational capabilities of HopeFund are partitioned into 14 distinct functional domains:

Operational Domain	Requirement Count	Functional Scope and Architectural Purpose
1. Authentication & Security (FR-AUTH)	5 Requirements	Role-based authentication, bcrypt password hashing, session timeouts, and anti-CSRF protection.
2. NGO Onboarding Lifecycle (FR-NGO)	6 Requirements	Registration submission, document upload, mandatory rejection rationale, and operational state transitions.
3. NGO Profile Management (FR-PROF)	4 Requirements	Public profile authoring, mission statements, gallery management, and governance review.
4. Campaign Authoring & Policies (FR-CAMP)	6 Requirements	Campaign publishing, deadline tracking, Target-Stop vs. Overflow policies, and emergency pauses.
5. Donation Purpose Workflows (FR-PURP)	4 Requirements	Predefined donation categories, custom purpose proposals, and NGO adjudication.
6. Simulated Payment Engine (FR-PAY)	6 Requirements	Mock transaction processing (BR-PAY-02 > 0.00 max 2 decimals), ledger logging, and simulated PDF receipts.
7. Fund Allocation & Budgeting (FR-FUND)	5 Requirements	Segregation of general donations into approved project budgets with mathematical ceiling enforcement.
8. Private Expense Ledger (FR-EXP)	6 Requirements	Itemized expenditure recording, mandatory proof bill upload, and strict quarantine access restriction.
9. Public Transparency Portal (FR-PUB)	5 Requirements	Aggregated financial visualizations, campaign progress tracking, and mandatory BR-EXP-04 disclaimer.
10. Facility Visit Scheduling (FR-VISIT)	6 Requirements	Configurable visiting slots, visitor reservation requests, dynamic capacity hold, and NGO adjudication.
11. Grievance Redressal (FR-COMP)	6 Requirements	Dispute lodging against NGOs or campaigns, admin investigation, and transparent dispute archival.
12. Automated Notifications (FR-NOTIF)	5 Requirements	Event-driven system notifications for application status, donations, visits, and grievances.
13. Action Audit History (FR-HIST)	4 Requirements	Immutable, append-only forensic audit logging capturing actor IP, timestamp, entity ID, and delta changes.

Operational Domain	Requirement Count	Functional Scope and Architectural Purpose
14. Reporting & Analytics (FR-REP)	4 Requirements	Tabular financial summaries, CSV/PDF statement exports, and realistic demonstration seed data.

5.3 User Classes, Characteristics, and Personas

The platform serves four primary authenticated user roles and unauthenticated public visitors:

User Class / Persona	Operational Focus	Functional Capabilities and Technical Expertise
Platform Administrator	System Governance	Technical staff or regulatory overseer responsible for vetting NGO onboarding applications, adjudicating disputes, and inspecting the immutable audit log.
NGO Operator	Charity Operations	Single administrative credential set representing the registered NGO entity. Manages campaigns, allocates funds, uploads expense receipts, and manages visit slots.
Registered Donor	Philanthropic Giving	Individual donor who authenticates to track simulated donation histories, download PDF receipts, propose custom purposes, and schedule facility visits.
Public Visitor	Public Community	Unauthenticated guest browsing public NGO directories, inspecting campaign progress, viewing aggregated expense summaries, and initiating guest donations.
System Auditor	Forensic Compliance	Specialized administrative persona with read-only inspection access to the complete financial ledger, proof files, and system audit logs.

5.4 Operating Environment and Technical Stack

The software operates reliably across the following development and production baselines:

- Server Operating System: Windows 10/11 (Local Development/XAMPP) or Linux Ubuntu 22.04 LTS (Production Cloud Target).
- Web Server: Apache HTTP Server 2.4+ with mod_rewrite enabled.
- Programming Runtime: PHP 8.2+ with PDO, OpenSSL, GD, and cURL extensions enabled.
- Database Management System: SQLite 3.35+ (zero-config local baseline) or MySQL 8.0+ / MariaDB 10.6+ (relational production baseline).
- Client Browsers: Google Chrome 110+, Mozilla Firefox 110+, Apple Safari 16+, Microsoft Edge 110+ (Desktop and Mobile Viewports).
- Responsive Viewport Breakpoints: Mobile (375px to 767px), Tablet (768px to 1023px), Desktop (1024px to 1920px+).

5.5 Design and Implementation Constraints

The system architecture is strictly constrained by the following core project invariants:

- Constraint 1: Simulated Payments Only: Real payment gateways are strictly prohibited. The database records is_simulated = TRUE for all transactions.
- Constraint 2: Strictly Positive Donation Amounts (BR-PAY-02): All donations must be greater than INR 0.00 with at most two decimal places. No artificial minimum or maximum is imposed.
- Constraint 3: Public Expense Disclaimer (BR-EXP-04): Public expense portals must prominently display: 'Expense figures are self-reported by the NGO and have not been independently verified.'
- Constraint 4: Private Document Quarantine: Sensitive founding documents and expense proof bills must reside in a protected non-web-accessible directory.
- Constraint 5: Single NGO Credential: Each registered NGO entity operates under a single shared administrative account.
- Constraint 6: Relational 3NF Architecture: All transactional tables must adhere to Third Normal Form with relational foreign keys and check constraints.

6. System Features and Functional Requirements

This section details all 72 functional requirements organized across the 14 operational domains. Every requirement is assigned a permanent identifier, priority (MUST, SHOULD, COULD), testable inputs, processing logic, outputs, and acceptance criteria.

6.1 Authentication, Account Management & Security (FR-AUTH)

Req ID	Feature Name	Inputs	Processing Logic	Outputs	Acceptance Criteria	Priority
FR-AUTH-01	User Registration	Full name, email, password, role selection.	Validate unique email, enforce strong password policy (min 8 chars, mixed case, digit, symbol), hash with bcrypt (cost 12), create user record.	Redirect to login with verification prompt.	Acceptance: Registration fails if email exists or password complexity fails.	MUST
FR-AUTH-02	Secure Authentication	Email address, password credentials.	Verify existence, verify bcrypt hash, establish secure session, regenerate session ID to prevent fixation.	Authenticated dashboard view, role-specific session cookie.	Acceptance: 30-minute session inactivity timeout enforced.	MUST
FR-AUTH-03	Role-Based Access Control	Requested URL, active user session.	Enforce role permissions (Admin, NGO, Donor, Guest); intercept unauthorized requests.	Render requested page or throw HTTP 403 Forbidden.	Acceptance: NGO operator cannot access Admin endpoints under any circumstances.	MUST
FR-AUTH-04	Session Management & Logout	Authenticated user logout action.	Destroy server session data, clear session cookie, log logout event.	Redirect to public homepage with success message.	Acceptance: Back-button navigation cannot access cached authenticated screens.	MUST
FR-AUTH-05	Anti-CSRF Protection	HTTP POST/PUT/DELETE requests.	Validate cryptographically random CSRF token embedded in session against submitted form token.	Process request if token matches; abort with HTTP 419 if token invalid.	Acceptance: 100% of state-mutating requests validate CSRF tokens.	MUST

6.2 NGO Onboarding & Operational Lifecycle Management (FR-NGO)

Req ID	Feature Name	Inputs	Processing Logic	Outputs	Acceptance Criteria	Priority
FR-NGO-01	Onboarding Submission	NGO legal name, registration number, registration type, founding document PDF.	Validate mandatory fields, verify PDF format (max 10MB), store document in quarantined directory, set state to PENDING.	Unique application reference, email notification to Admin.	Acceptance: Document cannot be accessed via direct public URL.	MUST
FR-NGO-02	Administrative Review	Admin review action, approval disposition, optional reviewer notes.	Fetch quarantined document for authenticated Admin review, display verification checklist.	Admin adjudication screen with document preview.	Acceptance: Quarantined document preview rendered securely via streaming controller.	MUST
FR-NGO-03	Application Approval	Admin approval confirmation.	Set NGO status to ACTIVE, initialize NGO financial ledger, enable campaign creation, send approval email.	Success notification, NGO operator dashboard activated.	Acceptance: NGO cannot accept donations until status becomes ACTIVE.	MUST
FR-NGO-04	Mandatory Rejection Rationale	Admin rejection action, mandatory textual explanation.	Verify non-empty rejection rationale, update status to REJECTED, record rationale in audit history.	Notification to NGO applicant detailing specific deficiencies.	Acceptance: Rejection cannot be saved without an explicit explanatory note.	MUST
FR-NGO-05	Operational State Suspension	Admin suspension action, suspension reason.	Transition ACTIVE NGO to PAUSED status; lock active campaigns from receiving new donations.	Public profile displays 'Operations Temporarily Paused' banner.	Acceptance: Paused NGOs retain historical records; cannot receive donations.	MUST
FR-NGO-06	NGO Archival Governance	Admin archival action, governance notes.	Transition NGO to ARCHIVED status; permanently disable campaign creation and login.	Public profile marked 'Archived Organization'; historical data preserved.	Acceptance: Archival uses ON DELETE RESTRICT; zero cascade deletion.	MUST

6.3 NGO Profile Management (FR-PROF)

Req ID	Feature Name	Inputs	Processing Logic	Outputs	Acceptance Criteria	Priority
FR-PROF-01	Profile Authoring	Mission statement, vision, causes, contact details, logo image.	Sanitize textual inputs, validate image format (PNG/JPG max 2MB), update profile record.	Updated public profile preview.	Acceptance: Cross-site scripting (XSS) payloads stripped from all fields.	MUST
FR-PROF-02	Public NGO Directory	Search terms, cause filter, location filter.	Execute parameterized query across approved ACTIVE NGOs, paginate results (12 per page).	Grid view of verified NGO cards with search/filter controls.	Acceptance: Only ACTIVE NGOs appear in public directory search results.	MUST
FR-PROF-03	Bank Account Details (Demo)	Bank name, branch, account number, IFSC code.	Validate field formatting, record demo account details for administrative review.	Encrypted demo bank profile record.	Acceptance: Fields marked as simulated demo data.	SHOULD
FR-PROF-04	Impact Gallery Upload	Activity photos, caption, project tag.	Validate image dimensions, store compressed image in public media directory.	Public impact gallery slider on NGO profile page.	Acceptance: Maximum 10 active gallery photos per NGO.	COULD

6.4 Campaign Authoring & Lifecycle Policies (FR-CAMP)

Req ID	Feature Name	Inputs	Processing Logic	Outputs	Acceptance Criteria	Priority
FR-CAMP-01	Autonomous Campaign Creation	Title, target goal, end date, description, cover image, closure policy.	Validate title length, ensure target > 0, set status to ACTIVE, record policy (Target-Stop vs Overflow).	Public campaign page with progress bar and donation button.	Acceptance: NGO can publish campaigns autonomously without Admin pre-approval.	MUST
FR-CAMP-02	Target-Stop Policy Enforcement	Incoming donation matching or exceeding target.	When total donations equal target goal, automatically set campaign status to COMPLETED; disable donation form.	Status badge updates to 'Goal Achieved: Campaign Closed'.	Acceptance: Zero donations accepted once Target-Stop campaign completes.	MUST
FR-CAMP-	Overflow Policy	Incoming	Allow donation	Progress bar	Acceptance:	MU

Req ID	Feature Name	Inputs	Processing Logic	Outputs	Acceptance Criteria	Priority
03	Enforcement	donation exceeding target goal.	processing, update total collected, calculate overflow percentage above 100%.	shows overflow indicator (e.g., 115% Funded).	Donations continue until explicitly closed or deadline expires.	ST
FR-CAMP-04	Deadline Expiration Automation	System cron / query-time date check.	Compare current timestamp against campaign end_date; if expired, transition status to EXPIRED.	Donation button disabled with 'Campaign Concluded' message.	Acceptance: Expired campaigns reject new donation submissions unconditionally.	MUST
FR-CAMP-05	Emergency Campaign Pause	NGO operator pause action, pause reason.	Transition ACTIVE campaign to PAUSED status; temporarily disable donation intake.	Public campaign banner: 'Donations Temporarily Paused'.	Acceptance: NGO can resume paused campaign at any time before deadline.	SHOULD
FR-CAMP-06	Campaign Progress Visualization	Campaign ID request.	Compute percentage collected = (collected_amount / target_amount) * 100, fetch donor count.	Real-time visual progress bar and public impact metrics.	Acceptance: Precision displayed to two decimal places without floating-point error.	MUST

6.5 Donation Channels & Custom Purpose Workflows (FR-PURP)

Req ID	Feature Name	Inputs	Processing Logic	Outputs	Acceptance Criteria	Priority
FR-PURP-01	Predefined Purpose Selection	NGO selected purposes (e.g., Education, Medical, Disaster Relief).	Bind standardized purpose taxonomy to NGO profile for donor selection.	Configured purpose dropdown on NGO donation form.	Acceptance: Every NGO supports at least one active donation purpose.	MUST
FR-PURP-02	Custom Purpose Proposal	Donor proposed purpose title, description, intended amount.	Record proposal in donation_purposes with status = PENDING_APPROVAL; link to NGO.	Proposal submitted notification to NGO dashboard.	Acceptance: Custom purpose cannot receive donations until approved by	MUST

Req ID	Feature Name	Inputs	Processing Logic	Outputs	Acceptance Criteria	Priority
					NGO.	
FR-PURP-03	Custom Purpose Adjudication	NGO operator review action (Approve/Reject), adjudication notes.	Update status to APPROVED or REJECTED; if approved, activate purpose channel.	Status notification to donor; purpose appears in NGO donation form.	Acceptance: Rejected purpose permanently locked with explanatory note.	MUST
FR-PURP-04	Purpose Fund Segregation	Simulated donation referencing purpose.	Tag donation record with purpose_id; aggregate purpose totals in ledger.	Itemized purpose balance reflected in NGO financial ledger.	Acceptance: Sum of purpose collections exactly reconciles to general ledger.	MUST

6.6 Simulated Payment Engine & Tamper-Evident Receipts (FR-PAY)

Req ID	Feature Name	Inputs	Processing Logic	Outputs	Acceptance Criteria	Priority
FR-PAY-01	Simulated Payment Checkout	Donation amount, donor name, email, payment method (Mock UPI/Card/NetBanking).	Validate input fields, verify campaign/NGO active status, display mock payment gateway modal.	Interactive mock payment interface with instant confirmation.	Acceptance: No real payment gateway calls; is_simulated flag set to TRUE.	MUST
FR-PAY-02	Donation Amount Validation (BR-PAY-02)	Donation amount field value.	Enforce rule BR-PAY-02: Amount A MUST be greater than INR 0.00 with at most two decimal places. Reject zero, negative, or non-numeric inputs. Zero artificial minimum or maximum.	Inline validation pass or error message: 'Please enter a valid positive donation amount greater than INR 0.00'.	Acceptance: Accepts INR 0.01, INR 1.00, INR 500.50; strictly rejects 0.00, -10.00, and abc.	MUST
FR-PAY-03	Transaction Processing & Ledger Entry	Mock payment approval trigger.	Begin ACID transaction: insert donation record, increment campaign collected_amount, update NGO ledger, log	Transaction success confirmation screen with reference	Acceptance: All ledger mutations rollback atomically on	MUST

Req ID	Feature Name	Inputs	Processing Logic	Outputs	Acceptance Criteria	Priority
			audit record, commit transaction.	code.	any database failure.	
FR-PAY-04	Tamper-Evident PDF Receipt	Completed donation ID.	Generate verifiable PDF receipt with SHA-256 verification hash, transaction ID, date, amount, donor name, and NGO details.	Downloadable simulated PDF receipt document.	Acceptance: Prominently displays legal non-tax disclaimer per BR-PAY-05.	MUST
FR-PAY-05	Anonymous Donation Support	Donor checkbox: 'Keep my donation anonymous'.	Set is_anonymous = TRUE; mask donor name on public campaign donor lists while retaining audit record.	Public list shows 'Anonymous Donor'; donor retains receipt access.	Acceptance: Anonymous donor identity hidden from public views.	MUST
FR-PAY-06	Donor Giving History	Authenticated donor dashboard request.	Query all donations matching donor_id, calculate lifetime contributions, display downloadable receipt links.	Tabular giving history with status, date, NGO, and PDF links.	Acceptance: Donor sees complete personal transaction history.	MUST

6.7 Fund Allocation & Project Budgeting (FR-FUND)

Req ID	Feature Name	Inputs	Processing Logic	Outputs	Acceptance Criteria	Priority
FR-FUND-01	Project Budget Definition	Project title, allocated amount, target outcomes, timeline.	Verify allocated amount <= available unallocated NGO funds (BR-FUND-02), create project record.	New project initialized with authorized budget ceiling.	Acceptance: Allocation cannot exceed available unallocated ledger balance.	MUST
FR-FUND-02	Fund Allocation Transfer	Source general funds, target project ID, allocation amount.	Deduct from unallocated ledger balance, credit project allocated_budget, record allocation audit entry.	Updated project financial summary and ledger state.	Acceptance: Conservation of funds invariant maintained: Unallocated + Allocated = Total.	MUST
FR-FUND-03	Project Status Lifecycle	NGO status update	Validate state transition rules; freeze budget	Updated project status	Acceptance: Closed projects	MUST

Req ID	Feature Name	Inputs	Processing Logic	Outputs	Acceptance Criteria	Priority
		(Planning, In-Progress, Completed, Closed).	changes if status is Closed.	indicator on public profile.	reject additional fund allocations.	
FR-FUND-04	Allocation History Log	Project ID financial inquiry.	Fetch chronological list of allocation events with timestamps and allocating user.	Detailed allocation audit table.	Acceptance: Every rupee allocated is traceable to a specific donation source or general pool.	SHOULD
FR-FUND-05	Budget Variance Alert	System calculation during expense entry.	Check if recorded expenses exceed 90% of allocated budget; trigger warning.	Advisory banner on NGO project dashboard: 'Budget 90% Utilized'.	Acceptance: Warning rendered without blocking valid remaining expenses.	COULD

6.8 Private Expense Ledger & Proof Verification (FR-EXP)

Req ID	Feature Name	Inputs	Processing Logic	Outputs	Acceptance Criteria	Priority
FR-EXP-01	Expense Entry Recording	Project ID, expense category, amount, payment date, vendor name, proof file.	Verify amount > 0, verify cumulative expenses <= project budget (BR-EXP-01), insert expense record.	Expense entry logged with status = RECORDED.	Acceptance: Cannot record expenses exceeding authorized project budget.	MUST
FR-EXP-02	Mandatory Expense Bill Upload (BR-EXP-02)	Expense bill file (PDF, JPG, PNG max 5MB).	Validate file MIME type, sanitize filename, store file in quarantined directory, record file hash.	Secure proof document record linked to expense ID.	Acceptance: Expense entry strictly rejected without valid uploaded proof file.	MUST
FR-EXP-03	Expense Ledger Privacy Wall (BR-EXP-03)	Requested expense view endpoint, user role.	Enforce privacy wall: Allow access only to authenticated NGO operators of that organization and Admin.	Detailed itemized expense table and bill preview links.	Acceptance: Public visitors and donors strictly denied access to itemized	MUST

Req ID	Feature Name	Inputs	Processing Logic	Outputs	Acceptance Criteria	Priority
					expenses.	
FR-EXP-04	Administrative Expense Audit	Admin inspection action.	Admin inspects expense details, views proof bills, adds audit verification notes.	Audit flag: 'Verified by Platform Administrator'.	Acceptance: Admin audit trail recorded in system audit log.	SHOULD
FR-EXP-05	Proof Document Quarantine	Direct URL request to proof bill path.	Web server blocks direct file access (.htaccess / location block); requests routed through secure streaming controller.	Stream file only if user has authorized session; else HTTP 403.	Acceptance: 100% of proof files unreachable via direct URL browsing.	MUST
FR-EXP-06	Expense Category Tagging	Standard category selection (Supplies, Logistics, Personnel, Direct Aid).	Tag expense with category; aggregate category totals for public reporting.	Category tagged record.	Acceptance: Uncategorized expenses rejected at validation gate.	MUST

6.9 Public Transparency Portal & Aggregate Reporting (FR-PUB)

Req ID	Feature Name	Inputs	Processing Logic	Outputs	Acceptance Criteria	Priority
FR-PUB-01	Aggregate Expense Visualization	Public NGO profile request.	Query sum of expenses grouped by category; calculate percentages; mask individual vendor and bill details.	High-level category breakdown chart (e.g., Supplies 45%, Logistics 30%).	Acceptance: Zero itemized bills or vendor names exposed to the public.	MUST
FR-PUB-02	Public Expense Disclaimer (BR-EXP-04)	Rendering of public expense summaries.	Unconditionally render exact disclaimer text: 'Expense figures are self-reported by the NGO and have not been independently verified.'	Prominent disclaimer banner positioned directly above public expense charts.	Acceptance: Disclaimer text displayed word-for-word on all public expense views.	MUST
FR-PUB-03	Fund Utilization Ratio	Public NGO profile request.	Compute utilization = (total_expenses /	Visual utilization	Acceptance: Displayed with	SHOULD

Req ID	Feature Name	Inputs	Processing Logic	Outputs	Acceptance Criteria	Priority
			total_donations_collected) * 100.	metric widget.	explicit clarification that figures reflect self-reported data.	
FR-PUB-04	Public Project Progress Feed	Public project ID request.	Fetch approved project milestones, narrative updates, and non-sensitive photos.	Chronological project progress feed.	Acceptance: Milestones visible to public donors to track philanthropic impact.	SHOULD
FR-PUB-05	Global Platform Metrics	Homepage request.	Aggregate platform-wide totals: verified NGOs, active campaigns, simulated donations, visitors accommodated.	High-level summary counters on landing page.	Acceptance: Counters reflect real-time database aggregates.	COULD

6.10 Facility Visit Scheduling & Capacity Management (FR-VISIT)

Req ID	Feature Name	Inputs	Processing Logic	Outputs	Acceptance Criteria	Priority
FR-VISIT-01	Visit Slot Configuration	Date, start time, end time, maximum visitor capacity (e.g., 10 seats).	Validate date >= today, ensure start < end, insert visit_slots record with available_capacity = max_capacity.	Available visit slot published on NGO public calendar.	Acceptance: Maximum capacity must be a positive integer >= 1.	MUST
FR-VISIT-02	Visitor Slot Reservation	Slot ID, visitor name, visitor count, contact email, purpose.	Check available_capacity >= visitor_count; decrement available_capacity; insert booking with status = PENDING.	Booking confirmation with unique reference; pending NGO review.	Acceptance: System rejects booking if requested count exceeds available capacity.	MUST
FR-VISIT-03	Mandatory NGO Adjudication (BR-VISIT-01)	NGO review action (Approve/Reject), optional review note.	Update booking status to CONFIRMED or REJECTED; send email notification to visitor.	Updated booking status on visitor and NGO dashboards.	Acceptance: Visits must be explicitly confirmed by NGO before visitor arrival.	MUST
FR-VISIT-04	Dynamic	Visitor	Update booking status	Slot available	Acceptance:	MU

Req ID	Feature Name	Inputs	Processing Logic	Outputs	Acceptance Criteria	Priority
	Capacity Restoration (BR-VISIT-02)	cancellation or NGO rejection.	to CANCELLED or REJECTED; atomically restore visitor_count back to slot available_capacity.	capacity incremented immediately.	Released seats become instantly bookable by other visitors.	ST
FR-VISIT-05	Visit Rescheduling Workflow (BR-VISIT-03)	Rescheduling request, new target slot ID.	Verify capacity in target slot; release capacity from old slot; transfer booking to new slot with status = PENDING.	Reschedule confirmation; awaiting NGO re-adjudication.	Acceptance: Cannot reschedule to a fully booked slot.	SHOULD
FR-VISIT-06	Visitor Attendance Verification	NGO check-in action on visit date.	Mark booking status = COMPLETED; record verified attendee count.	Attendance marked in visit ledger.	Acceptance: Historical attendance tracked for institutional transparency.	COULD

6.11 Grievance Redressal / Complaint Management (FR-COMP)

Req ID	Feature Name	Inputs	Processing Logic	Outputs	Acceptance Criteria	Priority
FR-COMP-01	Non-Disruptive Complaint Lodging (BR-COMP-01)	Target NGO/Campaign ID, complainant email, category, detailed grievance narrative.	Insert complaints record with status = SUBMITTED; generate complaint reference; do not pause NGO operations.	Submission acknowledgment with tracking code; notification to Admin.	Acceptance: Complaint lodging does not automatically freeze NGO fundraising.	MUST
FR-COMP-02	Administrative Investigation	Admin complaint review, inquiry notes, status transition to UNDER_INVESTIGATION.	Record investigation milestones in dispute audit log; request clarification from NGO.	Dispute tracking view updated.	Acceptance: Complainant and NGO can view non-confidential investigation status.	MUST
FR-COMP-03	Mandatory Adjudication Rationale (BR-COMP-02)	Admin resolution action (Upheld, Dismissed), mandatory adjudication rationale.	Verify non-empty resolution rationale; update status to RESOLVED; execute corrective action if upheld.	Formal resolution notice sent to complainant and NGO.	Acceptance: Admin cannot close a complaint without documented justification.	MUST

Req ID	Feature Name	Inputs	Processing Logic	Outputs	Acceptance Criteria	Priority
FR-COMP-04	Sanction Execution Workflow	Admin finding: Upheld with sanctions.	If severe violation confirmed, execute manual NGO suspension or campaign freeze per FR-NGO-05.	NGO operational status updated; logged in audit history.	Acceptance: Sanctions require explicit Admin confirmation and audit record.	SHOULD
FR-COMP-05	Permanent Dispute Record Archival (BR-COMP-03)	Resolution closure.	Archive dispute records permanently; prevent deletion of complaint history.	Permanent audit record retained for regulatory compliance.	Acceptance: Complaints cannot be deleted from the database under any role.	MUST
FR-COMP-06	Complainant Dispute Status Tracker	Complaint reference code, complainant email.	Query complaints table and display current progress, resolution notes, and closure date.	Public tracking portal with read-only progress indicator.	Acceptance: Complainant can track dispute without requiring an account.	SHOULD

6.12 Automated Notifications (FR-NOTIF)

Req ID	Feature Name	Inputs	Processing Logic	Outputs	Acceptance Criteria	Priority
FR-NOTIF-01	Lifecycle Event Triggers	System events (NGO Approved/Rejected, Donation Received, Visit Booked, Complaint Lodged).	Construct notification message, insert into notifications table with recipient user_id and read_status = FALSE.	Unread notification badge incremented on recipient dashboard.	Acceptance: Notifications generated asynchronously within 500ms of event.	MUST
FR-NOTIF-02	In-App Notification Feed	Authenticated user notification panel request.	Fetch notifications for current user ordered by created_at DESC; paginate (15 per view).	Dropdown notification list with timestamps and action links.	Acceptance: User can mark individual or all notifications as read.	MUST
FR-NOTIF-03	Simulated Email Dispatch	Recipient email, subject, message body.	Log outgoing email payload to application log file (logs/simulated_emails.	Simulated email logged for test verification	Acceptance: 100% of critical lifecycle events generate	SHOULD

Req ID	Feature Name	Inputs	Processing Logic	Outputs	Acceptance Criteria	Priority
			log) with full headers.	without external SMTP dependency.	simulated emails.	
FR-NOTIF-04	Notification Preferences	User notification toggle settings.	Persist user preferences for optional notification categories.	Updated preferences profile.	Acceptance: Mandatory administrative alerts cannot be disabled.	COULD
FR-NOTIF-05	Urgent Banner Broadcast	Admin global announcement text.	Publish platform-wide broadcast banner displayed to all authenticated users.	Dismissible announcement banner across top of UI.	Acceptance: Displayed across all active user sessions.	COULD

6.13 Action Audit History (FR-HIST)

Req ID	Feature Name	Inputs	Processing Logic	Outputs	Acceptance Criteria	Priority
FR-HIST-01	Immutable Audit Logging (BR-HIST-01)	System state mutation (Create, Update, Delete, Adjudicate, Allocate, Login).	Capture actor user_id, role, client IP address, action_type, entity_name, entity_id, delta payload, timestamp; insert into action_history.	Append-only audit record committed in same ACID transaction.	Acceptance: No UPDATE or DELETE SQL statements exist for action_history table.	MUST
FR-HIST-02	Administrative Audit Inspection	Admin audit search parameters (Date range, actor, action type, entity).	Execute indexed query across action_history; render tabular audit log with JSON delta diff viewer.	Filterable forensic audit view with timestamp precision to the second.	Acceptance: Accessible exclusively to authenticated Platform Administrators.	MUST
FR-HIST-03	Tamper-Evident Record Chaining	Consecutive audit log entries.	Compute SHA-256 hash of previous log entry and current record fields, creating a tamper-evident audit hash chain.	Cryptographic hash stored in audit_hash column.	Acceptance: Any manual database tampering breaks cryptographic verification chain.	SHOULD
FR-HIST-04	Forensic Audit	Admin export	Extract filtered audit log	Downloadabl	Acceptance:	COU

Req ID	Feature Name	Inputs	Processing Logic	Outputs	Acceptance Criteria	Priority
	Log Export	request (CSV / JSON format).	slice into structured CSV or JSON file.	e forensic audit report file.	Export includes complete cryptographic metadata and timestamps.	LD

6.14 Reporting, Analytics & Demonstration Seed Data (FR-REP)

Req ID	Feature Name	Inputs	Processing Logic	Outputs	Acceptance Criteria	Priority
FR-REP-01	NGO Financial Statement Export	NGO operator report request, date range filter.	Aggregate total donations, allocations, itemized expenses, and remaining balances; format as structured tabular statement.	Downloadable financial summary (CSV / PDF format).	Acceptance: Reconciles 100% to internal ledger mathematical invariants.	MUST
FR-REP-02	Administrative Overview Analytics	Admin dashboard request.	Compute platform KPI metrics: total onboarded NGOs, active campaigns, simulated donations, visitor counts, open grievances.	Visual analytics dashboard with trend charts.	Acceptance: Dashboard loads in < 1.0 second under baseline dataset.	SHOULD
FR-REP-03	Comprehensive Seed Data Suite	System seeding script execution.	Populate database with realistic, internally consistent demo data: 4 NGOs across distinct states, 8 campaigns, 30+ donations, expenses with proofs, visit slots, and grievances.	Fully functional platform populated for evaluation.	Acceptance: Seed data adheres 100% to all 34 business rules and ledger invariants.	MUST
FR-REP-04	Donor Impact Summary Report	Donor annual impact statement request.	Aggregate all contributions made by donor across calendar year, broken down by NGO and cause.	Personalized annual donor impact summary statement.	Acceptance: Explicitly designated as a simulated educational summary.	COULD

7. External Interface Requirements

7.1 User Interfaces (UI)

HopeFund delivers a clean, responsive, accessible web interface adhering to modern user experience engineering principles:

- Responsive Design Matrix: Mobile-first responsive grid adapting seamlessly across mobile (375px+), tablet (768px+), and desktop (1024px to 1920px+) viewports.
- Executive Color Scheme: Professional, high-contrast academic palette utilizing Deep Navy (#1F4E79), Slate Blue (#2F5597), Charcoal (#333333), and Coral/Amber status accents.
- Accessibility Standard: Full compliance with W3C WCAG 2.1 Level AA, ensuring minimum color contrast ratios of 4.5:1 for normal text and 3:1 for large text, keyboard navigability, and screen reader semantic ARIA tags.
- Intuitive Form Validation: Real-time inline feedback for input formatting, field boundary rules, and file upload restrictions, preventing submission errors before network round-trips.

7.2 Hardware Interfaces

The software specifies the following minimal and recommended hardware baselines:

Hardware Deployment Tier	Processor Benchmark	Memory Requirement	Storage & Display Specifications
Server (Local Demonstration)	Dual-Core x86_64 CPU (2.0 GHz+)	4 GB RAM	500 MB free disk space for application files and demo uploads.
Server (Production Cloud)	Quad-Core virtual CPU (2.5 GHz+)	8 GB RAM	20 GB SSD storage with automated snapshot backups.
Client Workstation / Laptop	Modern x86_64 or ARM processor	2 GB available RAM	Standard 1080p display (1920x1080) or 720p minimum.
Client Mobile Device	Modern smartphone (iOS 15+ / Android 10+)	2 GB RAM	Minimum viewport width: 375 CSS pixels.

7.3 Software and Service Interfaces

External software runtimes and libraries utilized by HopeFund are cataloged below:

Software Component	Supported Versions / Baselines	Interface Purpose and Functional Role
Operating System	Windows 10/11 or Ubuntu Linux 22.04 LTS	Provides OS execution environment, process management, and local file storage.
Web Server Engine	Apache HTTP Server 2.4+ / Nginx 1.20+	Handles HTTP/HTTPS protocol termination, virtual hosting, and URL rewriting via mod_rewrite.
Programming Runtime	PHP 8.2+ Engine (Core &	Executes application logic; requires PDO, pdo_sqlite,

Software Component	Supported Versions / Baselines	Interface Purpose and Functional Role
	Extensions)	pdo_mysql, openssl, gd, mbstring extensions.
Database Management	SQLite 3.35+ or MySQL 8.0+ / MariaDB 10.6+	Provides ACID-compliant relational data storage, foreign key constraints, and indexing.
PDF Generation Engine	FPDF / TCPDF / Dompdf Library	Renders server-side simulated PDF receipts and financial statements with SHA-256 verification hashes.
Web Browsers	Chrome, Firefox, Safari, Edge (Evergreen)	Renders client-side HTML5, CSS3, and JavaScript conforming to ECMAScript 2020 standards.

7.4 Communications Interfaces

The communication protocols governing network interactions include:

- Application Protocol: Hypertext Transfer Protocol Secure (HTTPS) over TLS 1.3 for all production client-server transactions; standard HTTP/1.1 supported for local isolated demonstration.
- Data Formatting: Client-server asynchronous data exchange utilizes JavaScript Object Notation (JSON) payloads with UTF-8 character encoding.
- File Upload Protocol: Multipart form-data (RFC 7578) with strict MIME-type validation and server-side file stream verification.
- Simulated Email Interface: Outgoing email notifications are written directly to server log streams (logs/simulated_emails.log) to avoid third-party SMTP dependencies.

8. Non-Functional Requirements

HopeFund defines 24 rigorous quality benchmarks structured across the 7 ISO/IEC 25010:2011 quality model characteristics. Every requirement specifies quantitative metrics and precise verification procedures.

8.1 Performance Requirements (NFR-PERF)

Req ID	Performance Benchmark	Quantitative Metric / Benchmark Target	Verification Procedure
NFR-PERF-01	Page Response Latency	95% of server response times for standard read requests must be under 500 ms under baseline concurrent load.	Benchmarked via automated HTTP load tests with 20 simulated concurrent users.
NFR-PERF-02	Simulated Checkout Latency	Mock payment processing, ledger update, and receipt generation must complete within 1.5 seconds end-to-end.	Timing logs recorded across 100 consecutive simulated checkout runs.
NFR-PERF-03	Database Query Efficiency	No single relational query shall require more than 100 ms to execute under expected baseline datasets.	Database slow query log analysis and SQL EXPLAIN query plan evaluation.
NFR-PERF-04	Static Asset Delivery	Client-side CSS, JavaScript, and image assets must be minified, delivering total initial page payloads under 1.5 MB.	Lighthouse web audit and network transfer inspection.

8.2 Security Requirements (NFR-SEC)

Req ID	Security Benchmark	Quantitative Metric / Benchmark Target	Verification Procedure
NFR-SEC-01	Password Cryptography	User passwords must be hashed using bcrypt with a work factor cost parameter of 12.	Static code audit verifying password_hash(..., PASSWORD_BCRYPT, ['cost' => 12]).
NFR-SEC-02	Session Governance	Sessions must timeout after 30 minutes of inactivity; cookies configured with HttpOnly, Secure, and SameSite=Lax.	Session testing verifying automatic invalidation after 30 minutes.
NFR-SEC-03	Anti-CSRF Mitigation	All state-mutating requests (POST/PUT/DELETE) must validate cryptographically signed tokens.	Automated penetration tests attempting request execution without tokens (expect HTTP 419).
NFR-SEC-04	SQL Injection Prevention	100% of database interactions must utilize parameterized prepared statements via PDO.	Static code analysis scanning for string concatenation in SQL queries; zero violations allowed.
NFR-SEC-05	XSS Prevention	All user-generated text rendered in HTML	Penetration tests injecting

Req ID	Security Benchmark	Quantitative Metric / Benchmark Target	Verification Procedure
		must be escaped using htmlspecialchars(..., ENT_QUOTES, 'UTF-8').	<script> payloads into all input fields; zero script execution.
NFR-SEC-06	Document Quarantine Protection	Sensitive founding documents and proof bills must reside in a protected non-web-accessible directory.	Direct HTTP requests to quarantined file paths must return HTTP 403 or 404.

8.3 Reliability and Availability (NFR-REL)

Req ID	Reliability Benchmark	Quantitative Metric / Benchmark Target	Verification Procedure
NFR-REL-01	Platform Availability Target	Platform must maintain 99.5% operational uptime during designated evaluation periods.	Server uptime logging and health-check monitoring across operational window.
NFR-REL-02	ACID Transaction Integrity	100% of financial ledger mutations must execute within explicit ACID transactions.	Automated chaos testing injecting database aborts; zero partial ledger records created.
NFR-REL-03	Graceful Degradation	System must catch unhandled exceptions, log diagnostic details, and present friendly error pages.	Test invocation of deliberate application exceptions; verify standard 500 error view.

8.4 Usability and Accessibility (NFR-USE)

Req ID	Usability Benchmark	Quantitative Metric / Benchmark Target	Verification Procedure
NFR-USE-01	WCAG 2.1 AA Compliance	All text elements must satisfy color contrast ratios >= 4.5:1 (3:1 for large text).	Automated accessibility evaluation via axe-core and Lighthouse accessibility audit.
NFR-USE-02	Intuitive Error Feedback	100% of form input errors must provide clear, human-readable correction guidance.	Manual usability walkthrough across all form submission workflows.
NFR-USE-03	Keyboard Navigability	All interactive components must be fully navigable and operable using standard keyboard tabs.	Full keyboard-only walkthrough confirming logical tab focus and visual focus outlines.
NFR-USE-04	Simulated Giving Transparency	Checkout and receipts must unambiguously communicate simulated educational status.	Inspection confirming non-tax disclaimers on 100% of payment screens and PDF

Req ID	Usability Benchmark	Quantitative Metric / Benchmark Target	Verification Procedure
			receipts.

8.5 Maintainability Requirements (NFR-MNT)

Req ID	Maintainability Benchmark	Quantitative Metric / Benchmark Target	Verification Procedure
NFR-MNT-01	Architectural Separation	Codebase must adhere to modular MVC separation: zero raw SQL queries inside view templates.	Static architectural analysis verifying complete decoupling of data and presentation.
NFR-MNT-02	Automated Seeding Capability	A single command must initialize the complete database schema and populate all seed data.	Automated test executing seeding script from scratch; verify clean baseline in < 10 seconds.
NFR-MNT-03	Documentation Completeness	All public methods and controller endpoints must feature standardized docblocks.	Docblock coverage inspection verifying >= 90% codebase documentation ratio.

8.6 Scalability Requirements (NFR-SCL)

Req ID	Scalability Benchmark	Quantitative Metric / Benchmark Target	Verification Procedure
NFR-SCL-01	Data Indexing and Query Plans	All foreign key columns and frequently queried attributes must feature B-Tree indexes.	Database schema audit verifying indexes on all foreign keys and status columns.
NFR-SCL-02	Result Set Pagination	All public and administrative data tables must enforce pagination (default 10 to 25 rows per page).	Verification of SQL LIMIT/OFFSET or cursor pagination across all list endpoints.

8.7 Portability and Compatibility (NFR-CMP)

Req ID	Compatibility Benchmark	Quantitative Metric / Benchmark Target	Verification Procedure
NFR-CMP-01	Cross-Platform Execution	Application must execute without modification on Windows (XAMPP) and Ubuntu Linux.	Successful deployment and test execution on both Windows 11 and Ubuntu

Req ID	Compatibility Benchmark	Quantitative Metric / Benchmark Target	Verification Procedure
			22.04 LTS.
NFR-CMP-02	Cross-Browser Rendering	Application UI must render identically across Chrome, Firefox, Edge, and Safari.	Visual regression testing across all 4 major evergreen browser engines.

9. Data Requirements and Data Management

9.1 Conceptual and Logical Data Architecture

HopeFund organizes its persistence layer into 18 core relational entities normalized to Third Normal Form (3NF). The relational schema eliminates functional and transitive dependencies while enforcing strict referential integrity through foreign key constraints.

9.2 Complete Entity Catalog and Data Dictionary

The 18 relational entities, their primary keys, foreign keys, and descriptions are detailed below:

Entity Name	Primary Key	Foreign Keys	Data Structure and Operational Role
users	user_id (INT PK AUTO)	None	Core user identities, email (unique), password_hash, role enum (admin, ngo, donor).
donor_profiles	donor_id (INT PK AUTO)	user_id (FK -> users)	Donor profile details: display name, contact phone, anonymous preference.
ngos	ngo_id (INT PK AUTO)	user_id (FK -> users)	NGO organizational master: legal name, registration number, registration type, status enum.
ngo_documents	doc_id (INT PK AUTO)	ngo_id (FK -> ngos)	Quarantined legal founding certificates, document type, file path, SHA-256 hash.
campaigns	campaign_id (INT PK AUTO)	ngo_id (FK -> ngos)	Fundraising campaigns: title, target_amount, collected_amount, closure_policy, status enum.
projects	project_id (INT PK AUTO)	ngo_id (FK -> ngos)	Operational charity initiatives: title, allocated_budget, recorded_expenses, status enum.
donation_purposes	purpose_id (INT PK AUTO)	ngo_id (FK -> ngos)	Predefined and custom donation channels: title, description, approval status enum.
donations	donation_id (INT PK AUTO)	ngo_id, campaign_id, donor_id	Simulated financial transactions: amount, payment_method, is_simulated, transaction_ref.
pdf_receipts	receipt_id (INT PK AUTO)	donation_id (FK -> donations)	Tamper-evident simulated receipts: receipt_number, pdf_path, verification_hash.
fund_allocations	allocation_id (INT PK AUTO)	ngo_id, project_id (FKs)	Internal fund transfers: amount, allocation_date, notes, allocating_user_id.
expenses	expense_id (INT PK AUTO)	project_id, ngo_id (FKs)	Itemized expenditure records: category, amount, payment_date, vendor_name, status.
proof_documents	proof_id (INT PK AUTO)	expense_id (FK ->	Quarantined vendor bills/receipts: file_name,

Entity Name	Primary Key	Foreign Keys	Data Structure and Operational Role
		expenses)	file_path, file_size, SHA-256 hash.
ngo_ledgers	ledger_id (INT PK AUTO)	ngo_id (FK -> ngos, Unique)	Mathematical balance ledger: total_collected, total_allocated, total_spent, balance.
visit_schedules	schedule_id (INT PK AUTO)	ngo_id (FK -> ngos)	Master visiting schedule configurations: days_of_week, operational_hours, guidelines.
visit_slots	slot_id (INT PK AUTO)	ngo_id (FK -> ngos)	Discrete visiting slots: slot_date, start_time, end_time, max_capacity, available_capacity.
visit_bookings	booking_id (INT PK AUTO)	slot_id, donor_id (FKs)	Visitor reservations: visitor_count, booking_status enum, purpose, adjudication_notes.
complaints	complaint_id (INT PK AUTO)	ngo_id, campaign_id (FKs)	Grievance records: complainant_email, category, narrative, status enum, resolution_notes.
action_history	audit_id (INT PK AUTO)	user_id (FK -> users, Nullable)	Append-only immutable audit trail: actor_ip, action_type, entity_id, delta_json, timestamp.

9.3 Data Classification and Privacy Governance

To prevent data leakage, application data is classified into four security tiers:

- Tier 1: Public Open Data: NGO names, public campaigns, aggregated category expense charts, published visit slot dates, and global platform metrics.
- Tier 2: Internal Authenticated Data: Unaggregated campaign donor lists, in-app notification records, and user profile preferences.
- Tier 3: Confidential Financial Data: Granular itemized expenses, vendor payment amounts, donor transaction histories, and complaint narratives.
- Tier 4: Restricted / Quarantined Data: NGO legal founding registration certificates and vendor expense proof bills. Stored in non-web-accessible directory; streamed only via authenticated controller.

9.4 Data Retention, Privacy, and Sanitization Policies

All transactional financial records (donations, allocations, expenses, receipts) are permanently retained to maintain historical ledger integrity. Archiving an NGO or user account transitions its status flag to ARCHIVED while preserving relational records (ON DELETE RESTRICT). User passwords are never stored in plaintext and are permanently encrypted with bcrypt. Audit logs are append-only and immutable.

10. System Architecture and Major Components

10.1 Four-Tier Architecture Pattern

HopeFund is architected as a layered four-tier system adhering to strict separation of concerns:

- Presentation Tier (Views): HTML5, CSS3, JavaScript, and responsive view templates rendering UI components, charts, and forms.
- Application Tier (Controllers): HTTP routing controllers processing inbound requests, validating inputs, enforcing CSRF tokens, and dispatching to services.
- Domain Service Tier (Business Logic): Core business logic services executing financial ledger invariants, campaign lifecycle policies, and notification dispatch.
- Data Access Tier (Models & Database): PDO abstraction layer executing parameterized SQL queries against normalized SQLite/MySQL relational tables.

10.2 Subsystem Decomposition and Component Boundaries

The platform is decomposed into 6 primary subsystems:

- Subsystem 1: Identity & Access Management (IAM): Manages registration, authentication, bcrypt hashing, session cookies, and RBAC guards.
- Subsystem 2: NGO Governance & Operations: Mediates onboarding document uploads, profile authoring, bank details, and operational status transitions.
- Subsystem 3: Campaign & Purpose Management: Handles autonomous campaign publishing, overflow policies, custom purpose adjudication, and deadline checks.
- Subsystem 4: Financial Ledger & Payment Simulation: Executes simulated checkouts, atomic ledger updates, project budgeting, and receipt generation.
- Subsystem 5: Transparency & Community Engagement: Manages expense proofs, category aggregations, visit slot bookings, and grievance redressal.
- Subsystem 6: Forensic Audit & System Analytics: Captures append-only audit events, generates financial statements, and renders KPI metrics.

10.3 Inter-Component Communication and Transaction Boundaries

All inter-component communications occur via synchronized PHP method calls within single process contexts. Critical business events requiring multi-table mutations (such as processing a simulated donation or recording an itemized expense) are wrapped inside explicit database transaction boundaries (BEGIN TRANSACTION ... COMMIT / ROLLBACK), guaranteeing ACID transaction guarantees.

11. Business Rules and Validation Rules

11.1 Fund Ledger Mathematical Invariants

To guarantee mathematical consistency across all financial operations, HopeFund strictly enforces the following ledger invariants:

Equation 1: Invariant 1: Total Collected Balance: Total Donations = Sum of all successful simulated donations for that NGO.

Equation 2: Invariant 2: Conservation of Funds: Total Donations = Unallocated General Funds + Sum of all Project Allocated Budgets.

Equation 3: Invariant 3: Project Budget Ceiling: Cumulative Itemized Expenses for Project P <= Authorized Allocated Budget for Project P.

Equation 4: Invariant 4: Non-Negative Balances: Unallocated Balance >= 0.00 and Project Remaining Balance >= 0.00 at all times.

11.2 Authoritative 34 Business Rules Register

The authoritative business rules governing platform behavior are detailed in the register below:

Rule ID	Rule Name	Authoritative Specification	Enforcement Category	Priority
BR-NGO-01	Fundraising Authorization Gate	NGOs must achieve ACTIVE status via Admin approval before receiving donations.	Access Control	MUST
BR-NGO-02	Mandatory Rejection Rationale	Admin cannot reject an NGO application without providing a detailed explanatory note.	Compliance	MUST
BR-NGO-03	Re-Application Isolation	Rejected NGOs can re-apply; original rejected record remains immutable in audit history.	Data Integrity	MUST
BR-NGO-04	Governed Profile Edits	Critical profile changes (legal name, registration number) require re-verification.	Compliance	SHOULD
BR-NGO-05	Paused Operational State	Paused NGOs retain historical records; cannot receive new donations or publish campaigns.	State Guard	MUST
BR-NGO-06	Archived Operational State	Archived NGOs permanently locked; historical transaction records preserved via ON DELETE RESTRICT.	Data Integrity	MUST
BR-CAMP-01	Autonomous Campaign Publishing	Verified ACTIVE NGOs can publish campaigns autonomously without Admin pre-approval.	Operational	MUST
BR-CAMP-02	Mandatory Closure Policy	Campaign authoring requires explicit selection of Target-Stop or Overflow policy.	Data Validation	MUST
BR-CAMP-03	Target-Stop Closure Policy	Campaigns with Target-Stop automatically close upon reaching 100% of target goal.	State Guard	MUST
BR-CAMP-	Overflow Past Target	Campaigns with Overflow accept donations beyond 100%	Operational	MU

Rule ID	Rule Name	Authoritative Specification	Enforcement Category	Priority
04	Policy	until deadline or manual closure.		ST
BR-CAMP-05	Deadline-Based Expiration	Campaigns automatically transition to EXPIRED when current date exceeds end date.	State Guard	MUST
BR-PAY-01	Strictly Simulated Financial Engine	No real payment gateway integrations; database records is_simulated = TRUE for all entries.	Safety Guard	MUST
BR-PAY-02	Strictly Positive Donation Amount	Every donation amount A MUST satisfy A > 0.00 with at most two decimal places. Zero, negative, and non-numeric entries are strictly rejected. No artificial minimum or maximum is imposed.	Data Validation	MUST
BR-PAY-03	Collection Integrity on Transaction	Ledger and campaign totals update only upon confirmed successful simulated transaction.	Math Invariant	MUST
BR-PAY-04	Idempotent Transaction Processing	Duplicate payment submissions with identical transaction tokens are rejected.	Idempotency	MUST
BR-PAY-05	PDF Receipt Non-Tax Disclaimer	Simulated PDF receipts must prominently state: 'Simulated Receipt: Educational Non-Tax Document'.	Legal Guard	MUST
BR-PAY-06	Refunds Prohibited in Initial Scope	Simulated donations cannot be refunded; educational simulation assumes finality.	Scope Guard	MUST
BR-PURP-01	Predefined Donation Purposes	NGOs must configure at least one standardized donation purpose.	Operational	MUST
BR-PURP-02	Custom Purpose Approval Gate	Donor-proposed custom purposes require explicit NGO operator approval before activation.	Approval Gate	MUST
BR-FUND-01	Four-Tier Mathematical Segregation	Total Collected, Allocated, Expended, and Remaining Balances are strictly partitioned.	Math Invariant	MUST
BR-FUND-02	Project Allocation Ceiling	Funds allocated to a project cannot exceed the available unallocated general ledger balance.	Math Invariant	MUST
BR-EXP-01	Project Expense Ceiling Constraint	Cumulative recorded expenses for a project cannot exceed authorized project budget.	Math Invariant	MUST
BR-EXP-02	Mandatory Expense Proof Upload	Every logged expense entry requires an uploaded proof document (PDF/JPG/PNG max 5MB).	Compliance	MUST
BR-EXP-03	Expense Ledger Privacy Guard	Itemized expenses and uploaded bills are restricted to authorized NGO operators and Admin.	Security Guard	MUST
BR-EXP-04	Public Aggregation & Disclaimer Policy	Public portals display only aggregate category totals and remaining funds, accompanied by mandatory text: 'Expense figures are self-reported by the NGO and have not	Transparency	MUST

Rule ID	Rule Name	Authoritative Specification	Enforcement Category	Priority
		been independently verified.'		
BR-VISIT-01	Mandatory NGO Visit Adjudication	Visitor reservations must be explicitly confirmed by the NGO before the visit occurs.	Approval Gate	MUST
BR-VISIT-02	Slot Capacity Ceiling & Restoration	Available slot capacity cannot be exceeded; cancellations restore capacity dynamically.	Resource Guard	MUST
BR-VISIT-03	Rescheduling Workflow Discipline	Rescheduling releases capacity from source slot and requires capacity in target slot.	Operational	SHOULD
BR-COMP-01	Non-Disruptive Complaint Intake	Dispute lodging creates an investigation ticket without automatically freezing NGO operations.	Dispute Policy	MUST
BR-COMP-02	Mandatory Adjudication Rationale	Admin cannot resolve a grievance without documenting a detailed rationale.	Compliance	MUST
BR-COMP-03	Permanent Dispute Record Archival	Grievance records cannot be deleted; retained permanently in audit log.	Data Integrity	MUST
BR-SEC-01	Single Shared NGO Account	Each registered NGO operates under a single shared administrative credential set.	Identity Model	MUST
BR-SEC-02	Donor Tenant Privacy Isolation	Donors can inspect only their own personal profile, donations, and bookings.	Privacy Guard	MUST
BR-HIST-01	Immutable Action History Trail	Audit log records are append-only; cannot be updated or deleted by any user role.	Forensic Guard	MUST

11.3 Field Validation and Type Boundary Rules

The system enforces the following input constraints across all presentation forms:

- Donation Amount: Numeric float > 0.00 with max 2 decimal places (BR-PAY-02). Format: [0-9]+(\.[0-9]{1,2})?.
- Email Addresses: Valid RFC 5322 format, max 255 characters, strictly validated via filter_var(..., FILTER_VALIDATE_EMAIL).
- Passwords: Minimum 8 characters, at least 1 uppercase letter, 1 lowercase letter, 1 number, and 1 special symbol.
- Textual Inputs: Names (max 100 chars), Titles (max 150 chars), Descriptions (max 5,000 chars); sanitized against HTML injection.
- File Uploads: Founding documents (PDF only <= 10MB); Expense proof bills (PDF, JPG, PNG <= 5MB); MIME verified.

12. Error Handling and Exception Scenarios

12.1 Standardized Exception Taxonomy and Error Codes

HopeFund standardizes system error codes to facilitate debugging and auditing:

Error Code	Error Designation	Trigger Condition and Architectural Description	HTTP Status
ERR-AUTH-001	Invalid Credentials	Email or password does not match registered user records.	HTTP 401 Unauthorized
ERR-AUTH-002	Session Expired	Inactivity timeout exceeded 30 minutes; session terminated.	HTTP 401 Unauthorized
ERR-AUTH-003	Forbidden Role Access	User attempts to access an endpoint unauthorized for their role.	HTTP 403 Forbidden
ERR-VAL-001	Invalid Donation Amount	Donation amount <= 0.00 or exceeds 2 decimal places (BR-PAY-02).	HTTP 422 Unprocessable
ERR-VAL-002	Missing Mandatory Field	Required form field omitted or submitted with whitespace only.	HTTP 422 Unprocessable
ERR-VAL-003	CSRF Token Mismatch	Submitted form token missing, expired, or tampered.	HTTP 419 Page Expired
ERR-LEDGER-001	Exceeds Project Budget	Recorded expense amount exceeds authorized project budget balance.	HTTP 422 Unprocessable
ERR-LEDGER-002	Exceeds Unallocated Funds	Project allocation amount exceeds available NGO unallocated funds.	HTTP 422 Unprocessable
ERR-FILE-001	Invalid File Type	Uploaded file MIME type is not in approved list (PDF/JPG/PNG).	HTTP 415 Unsupported Media
ERR-FILE-002	File Size Exceeded	Uploaded file size exceeds maximum ceiling (5MB / 10MB).	HTTP 413 Payload Too Large
ERR-NOTFOUND-001	Resource Not Found	Requested NGO, campaign, or booking identifier does not exist.	HTTP 404 Not Found
ERR-RATE-001	Rate Limit Exceeded	More than 60 requests received from single IP within 60 seconds.	HTTP 429 Too Many Requests

12.2 User-Facing Notifications vs Diagnostic Logging

To prevent Information Disclosure vulnerabilities (STRIDE threat), the system separates user feedback from diagnostic error logs. User interfaces render sanitized, friendly error messages (e.g., 'Unable to process payment: please verify your amount and try again'). Full stack traces, database query strings, and system file paths are written exclusively to secure server logs (logs/error.log).

12.3 Automated Recovery Strategies and Transaction Rollback

In the event of an unexpected exception during a database operation, the active database transaction is rolled back immediately (PDO::rollBack()). Any newly uploaded temporary file is purged from disk, and the user session is returned to the pre-transaction state, preventing orphaned files or partial database records.

13. Use Cases and System Workflows

13.1 Use Case Catalog Overview

The 18 primary system use cases spanning Administrator, NGO Operator, Donor, and Public roles are cataloged below:

Use Case ID	Use Case Title	Primary Actor	Operational Scope and Goal
UC-ADM-01	Review NGO Onboarding Application	Platform Administrator	Adjudicate legal registration, founding PDF, and issue approval or rejection with mandatory rationale.
UC-ADM-02	Manage NGO Operational Lifecycle	Platform Administrator	Transition NGO operational states: Pause active operations or permanently Archive organizations.
UC-ADM-03	Investigate and Resolve Donor Grievances	Platform Administrator	Review lodged complaints, record investigation findings, and issue formal resolution rationale.
UC-ADM-04	Global Platform Audit & Oversight	Platform Administrator	Inspect immutable audit logs, verify transaction hash integrity, and monitor platform metrics.
UC-NGO-01	Profile Setup & Registration Maintenance	NGO Operator	Author public profile, mission statements, cause categories, and upload organization logo.
UC-NGO-02	Autonomous Campaign Authoring	NGO Operator	Publish targeted campaigns, configure Target-Stop vs Overflow policies, and manage campaign lifecycle.
UC-NGO-03	Review Custom Purpose Proposals	NGO Operator	Inspect donor-proposed charitable causes and approve or reject purpose channels.
UC-NGO-04	Allocate Collected Donations to Projects	NGO Operator	Transfer unallocated general funds into authorized project budgets with ceiling enforcement.
UC-NGO-05	Record Granular Expenses with Proof Bills	NGO Operator	Log itemized expenditures with category tagging and mandatory proof bill file uploads.
UC-NGO-06	Configure Visiting Schedules & Review Bookings	NGO Operator	Create visiting slots with capacity ceilings; confirm or reject pending visitor reservations.
UC-NGO-07	Generate and Export Financial Statements	NGO Operator	Export reconciled financial statements (CSV/PDF) adhering to ledger mathematical invariants.
UC-DNR-01	Discover Verified NGOs and Campaigns	Public Donor / Guest	Search and filter verified NGOs by cause and location; inspect campaign progress

Use Case ID	Use Case Title	Primary Actor	Operational Scope and Goal
			metrics.
UC-DNR-02	Authenticate and Manage Profile	Registered Donor	Register account, authenticate securely, and configure personal profile preferences.
UC-DNR-03	Simulate Financial Contribution	Public Donor / Guest	Execute simulated checkout with valid positive amount (BR-PAY-02) and obtain simulated receipt.
UC-DNR-04	Submit Custom Charitable Purpose	Registered Donor	Propose customized charitable purpose to a specific NGO for administrative review.
UC-DNR-05	Download Tamper-Evident Receipt	Public Donor / Guest	Download cryptographically verifiable PDF receipt with SHA-256 hash and non-tax disclaimer.
UC-DNR-06	Reserve Facility Visitor Slot	Public Donor / Guest	Book visitor reservation for an available NGO visiting slot with dynamic seat hold.
UC-DNR-07	Submit Formal Grievance or Dispute	Public Donor / Guest	Lodge dispute against an NGO or campaign with category tagging and tracking reference.

13.2 Detailed Behavioral Specifications for Key Primary Use Cases

UC-01: Public Donor Simulated Contribution & Instant Receipt

- Primary Actor: Public Donor / Unauthenticated Guest.
- Preconditions: Target NGO or Campaign is in ACTIVE operational status.
- Main Success Scenario:

1. Donor navigates to active Campaign or NGO profile and selects 'Donate Now'.

2. System renders donation form with simulated payment indicator and amount input.

3. Donor enters donation amount A > 0.00 (BR-PAY-02), donor name, email, and mock payment method.

4. System validates amount > 0.00 with max 2 decimal places; opens mock payment processing modal.

5. Donor clicks 'Authorize Simulated Payment'.

6. System starts ACID transaction: inserts donation record (is_simulated = TRUE), increments campaign collected_amount, updates NGO ledger, and logs audit record.

7. System commits transaction and generates tamper-evident simulated PDF receipt (BR-PAY-05).

8. System renders success screen with transaction reference code and receipt download button.

- Alternate Flow 3a: Non-Positive Amount: System rejects amount <= 0.00; renders error ERR-VAL-001; aborts.

- Postconditions: Donation logged in database; ledger updated; PDF receipt generated and available for download.

UC-02: NGO Campaign Creation with Configurable Overflow Policy

- Primary Actor: Authenticated NGO Operator.
- Preconditions: NGO is in ACTIVE verified status.
- Main Success Scenario:

1. NGO Operator navigates to Campaign Management and clicks 'Create New Campaign'.

2. System displays authoring form: title, target goal (INR), end date, description, cover image, closure policy.

3. Operator fills details, chooses Target-Stop or Overflow policy, and submits form.

4. System validates CSRF token, verifies target > 0 and end date >= today, and inserts campaign record with status = ACTIVE.

5. System logs campaign creation event in action_history.

6. System publishes campaign to public NGO directory immediately without Admin pre-approval (BR-CAMP-01).

- Postconditions: Campaign published; public can view and simulate donations.

UC-03: Private Expense Logging with Mandatory Proof Bill Upload

- Primary Actor: Authenticated NGO Operator.
- Preconditions: Project has an authorized allocated budget balance.
- Main Success Scenario:

1. Operator navigates to Project Expense Ledger and clicks 'Record New Expense'.

2. System presents expense form: category, amount, payment date, vendor name, bill upload.

3. Operator enters details and attaches vendor receipt (PDF/JPG/PNG <= 5MB).

4. System verifies: expense amount > 0, cumulative expenses <= project budget (BR-EXP-01), file MIME valid.

5. System moves uploaded bill to quarantined directory outside web root (BR-EXP-03).

6. System starts ACID transaction: inserts expense record, inserts proof_documents record, updates project spent total, updates NGO ledger spent total, logs audit record.

7. System commits transaction and renders updated private expense ledger.

- Alternate Flow 4a: Expense Exceeds Budget: System blocks entry with ERR-LEDGER-001; aborts transaction.
- Alternate Flow 4b: Missing Proof File: System blocks entry with BR-EXP-02 validation error; aborts.
- Postconditions: Expense logged; bill quarantined; project remaining balance decremented; category totals updated.

UC-04: Facility Visit Booking with Dynamic Capacity Release

- Primary Actor: Public Visitor / Registered Donor.
- Preconditions: NGO has configured active visit slots with available capacity.
- Main Success Scenario:

1. Visitor navigates to NGO profile and clicks 'Schedule a Visit'.

2. System displays calendar of available slots with remaining capacity.

3. Visitor selects slot, enters visitor count N, visitor name, email, and visit purpose.

4. System verifies available_capacity >= N; decrements available_capacity by N; inserts booking with status = PENDING.

5. System sends notification to NGO Operator for mandatory adjudication (BR-VISIT-01).

6. NGO Operator reviews booking and clicks 'Confirm Booking'.

7. System updates booking status to CONFIRMED; sends email notification to visitor.

- Alternate Flow 4a: Insufficient Capacity: System notifies visitor that seats are unavailable; prompts selection of another slot.
- Alternate Flow 6a: Booking Cancelled / Rejected: System updates status to CANCELLED/REJECTED and atomically restores N seats to slot available_capacity (BR-VISIT-02).
- Postconditions: Visit confirmed or seats dynamically restored.

UC-05: Administrative NGO Onboarding Adjudication

- Primary Actor: Platform Administrator.
- Preconditions: NGO applicant has submitted onboarding application with quarantined registration certificate.
- Main Success Scenario:

1. Administrator logs into Admin Dashboard and opens 'Pending NGO Applications'.

2. System displays application queue; Admin clicks an application to review.

3. System streams quarantined founding document securely through authenticated controller.

4. Admin verifies legal credentials, founding deed, and registration validity.

5. Admin clicks 'Approve Application'.

6. System starts ACID transaction: updates NGO status to ACTIVE, initializes ngo_ledgers record with zero balances, logs approval audit record.

7. System commits transaction and sends approval email notification to NGO.

- Alternate Flow 5a: Application Deficient: Admin clicks 'Reject Application'; enters mandatory explanatory rationale (BR-NGO-02); system updates status to REJECTED and notifies applicant.
- Postconditions: NGO is ACTIVE and capable of receiving donations, or REJECTED with documented rationale.

UC-06: Grievance Lodging and Dispute Resolution

- Primary Actor: Complainant (Donor / Public) and Platform Administrator.
- Preconditions: Targeted NGO or campaign exists in system database.
- Main Success Scenario:

1. Complainant navigates to 'File a Grievance' portal.

2. Complainant enters email, selects target NGO/campaign, chooses category, enters detailed narrative, and submits.

3. System validates fields, generates unique reference code, inserts complaint record with status = SUBMITTED.

4. System logs submission; does NOT pause NGO operations (BR-COMP-01 non-disruptive intake).

5. Administrator inspects complaint in Admin Grievance Management, sets status to UNDER_INVESTIGATION.

6. Admin investigates dispute, interviews NGO, inspects expense proofs, and reaches finding.

7. Admin enters mandatory resolution rationale (BR-COMP-02) and closes complaint as RESOLVED.

8. System sends resolution notification to complainant; permanently archives dispute record (BR-COMP-03).

- Postconditions: Grievance resolved and permanently archived in immutable audit log.

14. Requirements Traceability Matrix

This Requirements Traceability Matrix (RTM) establishes bidirectional traceability linking every functional domain, subsystem module, primary use case, test case identifier, and database entity.

Functional Domain	Subsystem Module	Traceable Use Cases	Traceable Test Cases	Relational Database Entities	Verification Method
FR-AUTH-01..05	Identity & Access Management	UC-DNR-02, UC-ADM-01	TC-AUTH-01..08	users, donor_profiles	Unit & Integration Test
FR-NGO-01..06	NGO Onboarding & Lifecycle	UC-ADM-01, UC-ADM-02, UC-NGO-01	TC-NGO-01..08	ngos, ngo_documents	Integration & E2E Test
FR-PROF-01..04	NGO Profile Management	UC-NGO-01, UC-DNR-01	TC-PROF-01..06	ngos	Integration & UI Test
FR-CAMP-01..06	Campaign Lifecycle Management	UC-NGO-02, UC-DNR-01	TC-CAMP-01..08	campaigns	Integration & Functional Test
FR-PURP-01..04	Donation Channels & Purposes	UC-NGO-03, UC-DNR-04	TC-PURP-01..06	donation_purposes	Integration & Business Logic Test
FR-PAY-01..06	Simulated Payment Engine (BR-PAY-02)	UC-DNR-03, UC-DNR-05	TC-PAY-01..10	donations, pdf_receipts, ngo_ledgers	ACID Transaction & Validation Test
FR-FUND-01..05	Fund Allocation & Budgeting	UC-NGO-04	TC-FUND-01..06	projects, fund_allocations, ngo_ledgers	Mathematical Invariant Test
FR-EXP-01..06	Private Expense Ledger & Proofs	UC-NGO-05	TC-EXP-01..08	expenses, proof_documents, projects	Security & Quarantine Test
FR-PUB-01..05	Public Transparency Portal (BR-EXP-04)	UC-DNR-01	TC-PUB-01..06	expenses, campaigns, ngos	UI & Disclaimer Verification Test
FR-VISIT-01..06	Facility Visit Scheduling	UC-NGO-06, UC-DNR-06	TC-VISIT-01..08	visit_schedules, visit_slots, visit_bookings	Concurrency & Seat Capacity Test
FR-COMP-01..06	Grievance Redressal / Complaints	UC-ADM-03, UC-DNR-07	TC-COMP-01..08	complaints	Lifecycle & Workflow Test
FR-NOTIF-01..05	Automated Event Notifications	All Use Cases	TC-NOTIF-01..06	notifications	Event Dispatch & Log Test
FR-HIST-01..04	Forensic Action Audit History	UC-ADM-04	TC-HIST-01..06	action_history	Immutability & Hash Chain Test

Functional Domain	Subsystem Module	Traceable Use Cases	Traceable Test Cases	Relational Database Entities	Verification Method
FR-REP-01..04	Reporting & Demonstration Data	UC-NGO-07, UC-ADM-04	TC-REP-01..06	ngo_ledgers, all tables	Reconciliation & Seeding Test

14.2 Non-Functional Requirements Traceability to Architectural Mechanisms

NFR Category	Quality Domain	Architectural Enforcement Mechanism and Control
NFR-PERF-01..04	Performance	B-Tree database indexes, minified assets, query limit/offset, efficient PDO connection pooling.
NFR-SEC-01..06	Security	Bcrypt hashing (cost 12), session timeouts, anti-CSRF tokens, parameterized SQL queries, document quarantine outside web root.
NFR-REL-01..03	Reliability	ACID transaction wrappers (PDO::beginTransaction), try-catch exception handling, graceful degradation.
NFR-USE-01..04	Usability	WCAG 2.1 AA compliant color palette, responsive CSS grid, inline validation, explicit simulation notices.
NFR-MNT-01..03	Maintainability	Strict MVC architectural separation, comprehensive docblocks, automated database seeding suite.
NFR-SCL-01..02	Scalability	B-Tree indexing on foreign keys and status flags, cursor-based pagination, stateless session architecture.
NFR-CMP-01..02	Compatibility	Standards-compliant HTML5/CSS3, cross-platform PHP runtime, support for all evergreen desktop and mobile browsers.

15. Acceptance Criteria

15.1 System-Wide Operational Acceptance Criteria

To achieve formal acceptance for academic capstone submission, HopeFund must satisfy the following criteria:

- 100% Simulation Fidelity: All financial donation flows are strictly simulated (is_simulated = TRUE); zero external banking APIs invoked.
- Strict Positive Donation Rule (BR-PAY-02): System accepts any valid positive amount > INR 0.00 with max 2 decimals; rejects 0.00, negative, and non-numeric inputs.
- Mandatory Public Expense Disclaimer (BR-EXP-04): Public expense charts display the exact disclaimer: 'Expense figures are self-reported by the NGO and have not been independently verified.'
- Zero Direct URL Access to Quarantined Files: Legal founding documents and expense bills cannot be opened via direct public browser URL.
- Mathematical Fund Ledger Integrity: Invariant Total Donations = Unallocated + Allocated is maintained across all transactions with zero floating-point error.
- Immutable Audit Trail: 100% of state-mutating actions produce immutable records in action_history; zero SQL DELETE or UPDATE operations permitted.

15.2 Module-Specific Verification and Sign-Off Matrix

Functional Module	Formal Verification and Acceptance Sign-Off Standard
Authentication & IAM	Enforces password complexity, bcrypt cost 12, 30-minute session expiration, and role separation between Admin, NGO, and Donor.
NGO Onboarding	Enforces mandatory founding document upload, document quarantine, and mandatory written rationale for all rejected applications.
Campaign Management	Enforces autonomous publishing, Target-Stop auto-closure at 100%, Overflow progression, and deadline expiration.
Simulated Donations	Enforces BR-PAY-02 (> 0.00 max 2 decimals), ACID ledger updates, and SHA-256 verifiable PDF receipts with legal disclaimers.
Expense Management	Enforces project budget ceiling, mandatory proof bill upload, and quarantine restriction to authenticated NGO/Admin users.
Facility Visits	Enforces slot capacity ceilings, mandatory NGO confirmation, and dynamic seat restoration upon cancellation/rejection.
Grievance Redressal	Enforces non-disruptive complaint intake, mandatory Admin resolution rationale, and permanent dispute archival.
Action Audit History	Enforces append-only immutable logging capturing actor, timestamp, IP, action, entity, and delta JSON.

16. Assumptions, Limitations, and Dependencies

16.1 Confirmed Strategic Baseline Assumptions (DEC-CONF-01..09)

Decision ID	Strategic Baseline Policy	Architectural Scope and Governing Principle
DEC-CONF-01	Simulation-Only Payment Architecture	Real payment gateway integrations (Razorpay, Stripe) are strictly prohibited; all payments are educational simulations.
DEC-CONF-02	Single Shared Credential per NGO	Each registered NGO operates under a single administrative user account; internal multi-staff delegation is deferred.
DEC-CONF-03	Non-Disruptive Complaint Intake	Filing a grievance creates an investigation ticket but does not automatically freeze NGO fundraising operations.
DEC-CONF-04	Private Document Quarantine Outside Web Root	Sensitive registration certificates and proof bills are stored in protected directories unreachable via direct URL browsing.
DEC-CONF-05	Strictly Positive Donation Amount (BR-PAY-02)	System accepts any valid positive amount > INR 0.00 with max 2 decimals; zero artificial minimum or maximum is imposed.
DEC-CONF-06	Public Expense Self-Reporting Disclaimer (BR-EXP-04)	Public portals display the exact disclaimer: 'Expense figures are self-reported by the NGO and have not been independently verified.'
DEC-CONF-07	Dynamic Facility Capacity Restoration	Cancellations or rejections of visit reservations immediately restore capacity to the corresponding visit slot.
DEC-CONF-08	Autonomous Campaign Authoring	Verified ACTIVE NGOs can publish fundraising campaigns autonomously without requiring Admin pre-approval.
DEC-CONF-09	Immutable Append-Only Audit Logging	Audit log entries cannot be modified or deleted; tamper-evident hash chaining guarantees historical audit integrity.

16.2 Known Project Scope Limitations

The current implementation baseline operates under the following explicit scope boundaries:

- No Real Currency Movement: Financial transactions are simulated; no interaction with banking networks, credit card processors, or UPI rails.
- No Automated Tax Certification: Generated PDF receipts are educational mock artifacts; platform does not issue legal Section 80G tax exemption certificates.
- Local File Storage Baseline: Quarantined documents and proof bills reside on the local server file system rather than distributed cloud object storage (AWS S3).
- Simulated Email Transport: Notifications are logged to server application streams rather than delivered through external production SMTP servers.

16.3 Technical Dependencies

- PHP Runtime: Requires PHP 8.2 or higher with PDO, pdo_sqlite, pdo_mysql, openssl, and gd extensions.

- Database Engine: Relational database engine supporting foreign key constraints and ACID transactions (SQLite 3.35+ or MySQL 8.0+).
- PDF Library: Server-side PDF generation utility (FPDF, TCPDF, or Dompdf) for dynamic receipt rendering.

17. References and Appendices

17.1 Formal Standards and Literature References

The engineering practices and requirements specifications in this document derive from authoritative standards:

Reference Tag	Formal Publication Details and Standard Citation
[IEEE 29148]	ISO/IEC/IEEE 29148:2018: Systems and software engineering: Life cycle processes: Requirements engineering. IEEE Computer Society, Piscataway, NJ.
[IEEE 830]	IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications. IEEE Computer Society, Piscataway, NJ.
[ISO 25010]	ISO/IEC 25010:2011: Systems and software engineering: Systems and software Quality Requirements and Evaluation (SQuaRE): System and software quality models. ISO, Geneva, Switzerland.
[OMG UML]	Object Management Group (OMG): Unified Modeling Language (OMG UML) Specification, Version 2.5.1, December 2017.
[IEEE 29119]	ISO/IEC/IEEE 29119:2013: Software and systems engineering: Software testing (Parts 1, 2, 3, and 4). IEEE Computer Society, Piscataway, NJ.
[IEEE 829]	IEEE Std 829-2008: IEEE Standard for Software and System Test Documentation. IEEE Computer Society, Piscataway, NJ.
[W3C WCAG]	World Wide Web Consortium (W3C): Web Content Accessibility Guidelines (WCAG) 2.1, W3C Recommendation, June 2018.

17.2 Appendix A: Hardware & Software Deployment Baseline

To ensure seamless evaluation by academic evaluators, HopeFund is packaged for zero-friction local execution under XAMPP. Place the project folder within htdocs, verify that Apache and MySQL/SQLite run under PHP 8.2+, and launch the local URL. The automated seeding utility populates the complete database schema and sample records in a single execution step.

17.3 Appendix B: Error Code and Message Catalog

A consolidated catalog of application exception codes, diagnostic meanings, and user feedback messages is maintained in the system source files under config/error_codes.php, providing end-to-end alignment with Section 12 of this specification.